

Course: B. Pharmacy

Subject Name: Pharmaceutical Organic Chemistry- II

Max Marks: 75

Date: 18/12/2024

Semester: III

Subject Code: BP301T

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1 Objective type of questions (Answer all the questions)

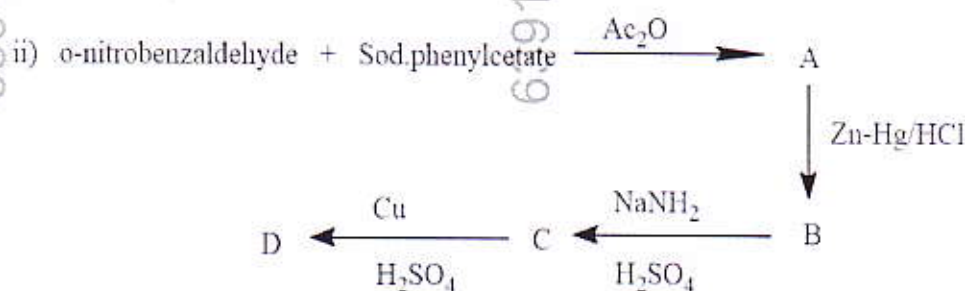
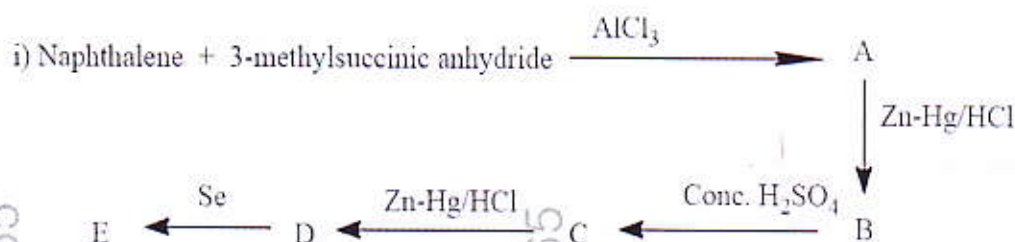
(10x2) = 20

- i) Explain the orbital picture of benzene and the type of hybridization involved.
- ii) Predict the product when benzoic acid reacts with PCl_5 and SOCl_2
- iii) How do fatty acids undergo esterification?
- iv) Write two qualitative tests used to identify phenols.
- v) What are the synthetic uses of aryl diazonium salts?
- vi) Draw and nomenclate 1, 4 dialin
- vii) Draw and nomenclate Anthraquinone
- viii) Draw and nomenclate α -phenyl-o-nitrocinnamic acid
- ix) Draw and nomenclate 9-acetylphenanthrene
- x) Draw and nomenclate 1,4 naphthaquinone

Q.2 Long answers (Answer 2 out of 3)

(10x2) = 20

- i) What are Analytical constants? Explain the principle and significance of any four analytical constants used in fat and oil analysis.
- ii) State Huckel's rule of aromaticity. Explain the aromaticity/non-aromaticity of Cyclopropenyl Cation, Cycloheptatrienyl Cation, Cyclobutadiene and Cyclooctatetraene.
- iii) Draw the structures and complete the following reaction by finding A, B, C, D, and E.



Q.3 Short Answers (Answer 7 out of 9)

(5x7) = 35

- i) Describe the mechanism of Friedel-Crafts alkylation and acylation with respect to the benzene, including the limitations of these reactions.
- ii) Discuss the effect of electron-donating and electron-withdrawing substituents on the reactivity and orientation of mono-substituted benzene compounds in electrophilic substitution reactions.
- iii) Explain the acidity of phenols, and how the presence of different substituents affects their acidity, give examples.
- iv) Explain the acidity of aromatic acids, the influence of substituents on its acidity, and some of the important reactions it undergoes.
- v) Explain the reaction, mechanisms of nitration, sulfonation, and halogenation of naphthalene. How do these reactions differ from those of benzene?
- vi) Describe the oxidation reactions of anthracene. Provide the mechanism for the formation of anthraquinone and explain its industrial significance.
- vii) Explain Baeyer's Strain Theory in the context of cycloalkanes and discuss its limitations. How does the theory fail to account for the observed stability of certain cycloalkanes?
- viii) Explain Structure and uses of DDT, Saccharin, BHC, Chloramine and Diphenylmethane.
- ix) Convert the following
 - a) Aniline to p-benzoquinone
 - b) Benzylamine to N-Benzylacetamide
 - c) Benzene to P-methylaniline
 - d) Benzoyl chloride to Benzamide
 - e) Toluene to Benzoic acid

****BEST OF LUCK****